

Agenda

1. Introduction & Properties of % wrt [+ , - , x, and Power]

2. Pair sum divisible by M

Starting 7:05

3. Find GCD(a, b)

A. Properties of GCD.

B. GCD

Modulo (%) Operator

$x = A \% B \rightarrow x$ is the remainder we will get when A is divided by B .

$$x = A \% B ; x = A - \text{Greatest multiple of } B \leq A.$$

$$\textcircled{1} 29 \% 6 = \underline{5}.$$

$$= 29 - \text{Greatest multiple of } 6$$

$$= 29 - 24 = 5 \leq 29$$

$$\textcircled{2} -3 \% 4 = \underline{1}.$$

$$-3 - \underline{\text{Greatest multiple of } 4}$$

$$\leq -3$$

$$= -3 - (-4)$$

$$= -3 + 4 = 1$$

Range of $A \% B$

$$= [0, B-1]$$

$$A = 5 ; B = 3 ;$$

$$\underline{[0, 2]}$$

$$5 \% 3 = 2 ; 7 \% 3 = 1$$

$$6 \% 3 = 0 \quad 8 \% 3 = 2$$

$$9 \% 3 = 0 ; 10 \% 3 = 1$$

Why do we need -1 operator?

$$\underline{-2 \times 10^9 \leq m \leq 2 \times 10^9}$$

$$\underline{-\infty \leq x \leq +\infty}$$

$\div m$

$$\downarrow$$
$$\underline{[0, m-1]}$$

Restrict my values
over a particular
range.

Modular Arithmetic

(+, -, *, /) %

$$1. (a + b) \% m = (a + m + b + m) \% m$$

$$a = 8; b = 9; m = 5$$

$$(8 + 9) \% 5 = (8 + 5 + 9 + 5) \% 5$$

$$17 \% 5 = (3 + 4) \% 5$$

$$= 7 \% 5$$

$$2 = 2.$$

$$2. (a * b) \% m = (a \% m * b \% m) \% m$$

$$a = 8; b = 9; m = 5$$

$$(8 * 9) \% 5 = (8 \% 5 * 9 \% 5) \% 5$$

$$72 \% 5 = (3 * 4) \% 5$$

$$2 = 12 \% 5 = 2$$

$$x \% y$$

$$x < y.$$

$$\begin{aligned}
 3. \quad (a+m) \% m &= (a + \overset{0}{m-1 \cdot m}) + 1 \cdot m \\
 &= \underbrace{(a + m)}_{[0, m-1]} + 1 \cdot m \rightarrow \text{redundant} \\
 &= (a + m)
 \end{aligned}$$

$x - y$

$$\begin{aligned}
 &[-2, 6] + 5 \\
 &\quad \downarrow \\
 &[-2+5, 6+5] \\
 &= [3, 11]
 \end{aligned}$$

$$\begin{aligned}
 4. \quad (a-b) \% m &= \underbrace{(a - \overset{0}{m-1} - b + m)}_{[-(m-1), (m-1)]} + 1 \cdot m \\
 &= \underbrace{(m - (m-1), m + (m-1))}_{[1, 2m-1]}
 \end{aligned}$$

$$a = 2; \quad b = 5; \quad m = 5$$

$$\text{L.H.S} = (2 - 5) + 5$$

$$= -3 - (-5)$$

$$= -3 + 5$$

$$= \underline{2}$$

$$(2 - 0 + 5) + 5$$

$$= (2 - 0 + 5) + 5$$

$$= (7 + 5)$$

$$= \underline{2}$$

$$a = 5; \quad b = 2; \quad m = 5$$

$$\begin{aligned} \text{L.H.S} &= (5 - 2) + 1.5 \\ &= \underline{\underline{3.5}} \end{aligned}$$

RHS

$$\begin{aligned} &= (5 + 1.5 - 2 + 1.5 \\ &\quad + 5) + 1.5 \\ &= (0 - 2 + 5) + 1.5 \\ &= \underline{\underline{3 + 1.5}} \\ &= \underline{\underline{3.5}} \end{aligned}$$

$$5. \quad a^b \% m = \underbrace{(a * a * a * \dots * a)}_{b \text{ times}} \% m$$

$$= (a \% m * a \% m * a \% m * \dots * a \% m) \% m$$

$$= (a \% m)^b \% m$$

Quiz :

$$\begin{aligned} \bullet \quad & \underset{\substack{\downarrow \\ a}}{(37^{103} - 1)} \% 12 \quad \underset{\substack{\downarrow \\ b}}{=} \quad = (a \cdot 12 - b \cdot 12 + 12) \cdot 12 \\ & = (37^{103} \cdot 12 - 1 \cdot 12 + 12) \cdot 12 \\ & = ((37 \cdot 12)^{103} - 1 + 12) \cdot 12 \\ & = (1^{103} - 1 + 12) \cdot 12 \\ & = (1 - 1 + 12) \cdot 12 \\ & = 12 \cdot 12 = 0 \end{aligned}$$

m is given.

< Question > : Given N array elements. Find the count of pairs (i, j) such that $(arr[i] + arr[j]) \% m = 0$.
($1 \leq N \leq 10^5$)

NOTE : $i \neq j$ and pair (i, j) is same as pair (j, i)

arr $\rightarrow$ [4 7 6 5 8 3]
 0 1 2 3 4 5

$m = 3$

(2, 3)
↓
(3, 2)

(0, 3) (1, 3) (2, 5)
(0, 4) (1, 4)

ans = 5.

Idea 1:- Consider every possible pair using 2 for loops.
T.C $\rightarrow O(N^2)$

Idea 2:- Use the property of modulo addition.

$$(a + b) \% m = 0 \quad = \quad (a \% m + b \% m) \% m$$

$[0, m-1] \quad \downarrow \quad [0, m-1]$

$[0, 2m-2]$

$[0, 2m-2]$.

Obs:- Everythg inside bracket should be a factor of m .

(2) There are only 2 factors of m in the range $[0, 2m-2]$.

ae: $0, m$.

example

33.

$[0, 2m]$ where $n = 33$

$[0, 64]$

so $-3m, -2m, -m, 0, m, 2m, 3m, 4m, \dots$

$$\frac{m=6}{a[i] + m} = \frac{(a[i] + m + a[j] + m) \cdot m}{a[i] + m}$$

5	→	0	0	0
3	→	1	1	$m-1$
6	→	2	2	$m-2$
4	→	3	3	$m-3$
5	→	4	⋮	⋮
6	→	5	$(m-1)$	1

$$ans = 5C_2 + 3 \times 6 + 6 \times 5 + 4C_2$$

$$(6, 12), (6, 15), (12, 18)$$

$$N=4$$

$$3+2+1$$

Example :

arr[] $\rightarrow$ [2 3 4 8 6 15 5 12 17 7 18 10 9 16 21]

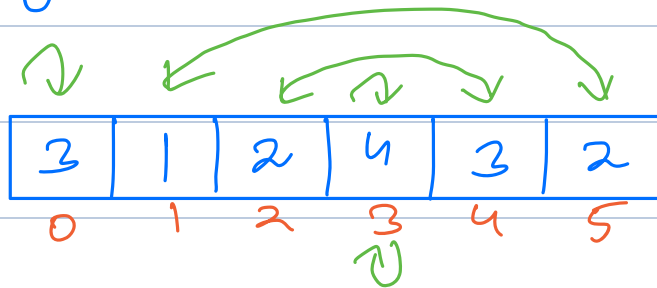
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

$$\frac{(N-1)(N)}{2}$$

$$M=6$$

$$N C_2$$

①. Cal. the val. of $(a C_2 \cdot m)$
 & store it in a freq.
 array.



$$3 C_2 + 1 \times 2 + 2 \times 3 + 4 C_2 \quad (4 \times 3/2)$$

$$\downarrow$$

$$3 \frac{(3-1)}{2} = 3 \times \frac{2}{2} = 3$$

Q:-
 SC $\rightarrow O(m)$ $\rightarrow$ because of frequency of array.
 T.C $\rightarrow \underline{O(N+m)}$.

Code

```
int freq[m];  $\forall i$  freq[i] = 0;
```

$\frac{N}{2}$ {
for (i = 0; i < N; i++) {
 val = a[i] + m;
 freq[val]++;
}

```
int ans = 0;  
ans += freq[0] * (freq[0] - 1) / 2;
```

if (m + 2 == 0) {
 ans += $\frac{\text{freq}[m/2] * (\text{freq}[m/2] - 1)}{2}$;
}

$m/2$ {
 l = 1; r = m - 1;
 while (l < r) {
 ans += freq[l] * freq[r];
 l++; r--;
 }
 return ans;

G.C.D (Greatest Common Divisor) ^(Highest Common Factor)

$\text{gcd}(A, B) = x$; x is the largest no. that satisfies

① $A \div x = 0$;

② $B \div x = 0$;

GCD (20, 65)

```

      ↓ ↓
      1 1
      2 5
      4 13
      5 65
      10
      20
  
```

GCD (-10, 20)

```

      ↓ ↓
     -10 1
      -5 2
     -2 4
      -1 5
       1 10
       2 20
       5
      10
  
```

GCD (7, 9)

```

      ↓ ↓
      1 1
      7 3
          9
  
```

GCD (0, 8)

```

      ↓ ↓
      1 1
      2 2
      3 4
      4 8
      8
      1
  
```

GCD (0, -10)

```

      ↓ -10
      1 -5
      2 -2
      3 1
      4 2
      5 5
      10
  
```

Or:-

GCD (4, 7)

```

      ↓ ↓
      1 1
      2 7
      4
  
```

Properties of GCD

$$(1) \quad \gcd(A, B) = \gcd(B, A)$$

Commutative Property

$$(2) \quad \begin{aligned} \gcd(A, B, C) &= \gcd(\gcd(A, B), C) \\ &= \gcd(\gcd(A, C), B) \\ &= \gcd(\gcd(B, C), A) \end{aligned}$$

Associative Property.

$$(3) \quad \gcd(0, A) = \underline{\text{abs}(A)}$$
$$= \underline{\underline{|A|}}.$$

④ Special Property:-

Given, $\gcd(A, B) = x$ $A, B > 0$ ~~KK~~ $A > B$



$$\gcd(A-B, B) = \underline{x}$$

① $(A-B) \div x = 0$

② $B \div x = 0$

③ x is the largest No.

Examples

①
$$\begin{aligned} (A-B) \div x &= \left(\underset{0}{A \div x} - \underset{0}{B \div x} + x \right) \div x \\ &= (0 - 0 + x) \div x \\ &= x \div x = \underline{0} \end{aligned}$$

①
$$\begin{aligned} \gcd(8, 6) &= 2 \\ &= \gcd(8-6, 6) = \gcd(2, 6) \\ &= \underline{2} \end{aligned}$$

$$A - B \geq B$$

$$\begin{aligned}
 \gcd(A, B) &= \gcd(A - B, B) \\
 &\quad \downarrow \\
 &= \gcd(A - 2B, B) \\
 &= \gcd(A - 3B, B) \\
 &\quad \vdots \\
 &= \gcd(A - xB, B) \\
 &= \underline{\gcd(A \bmod B, B)}
 \end{aligned}$$

$$\begin{aligned}
 \gcd(15, 6) &= \gcd(9, 6) \\
 &\quad \downarrow \\
 &= \gcd(15 \div 6, 6) \\
 &= \gcd(3, 6) \\
 &\quad \downarrow \\
 &\quad \quad \quad \swarrow \text{15} \div \underline{\underline{6}} \\
 &\quad \quad \quad \gcd(3, 6) \\
 &\quad \quad \quad \downarrow \\
 &\quad \quad \quad \gcd(6, 3) \\
 &\quad \quad \quad \downarrow \\
 &\quad \quad \quad \gcd(6 \div 3, 3) \\
 &\quad \quad \quad = \gcd(0, 3) = \underline{\underline{3}}
 \end{aligned}$$

$$\begin{aligned} \gcd(5, 4) &= \gcd(5-4, 4) \\ &\quad \downarrow \\ &= \gcd(1, 4) \\ &\quad \downarrow \\ &= \gcd(5+4, 4) \end{aligned}$$

$$\begin{aligned} \gcd(7, 4) &= \gcd(7-4, 4) \\ &\quad \downarrow \\ &= \gcd(3, 4) \\ &\quad \downarrow \\ &= \gcd(4, 3) \\ &\quad \downarrow \\ &= \gcd(4+3, 3) \\ &\quad \downarrow \\ &= \gcd(1, 3) \\ &\quad \downarrow \\ &= \gcd(3, 1) \\ &\quad \downarrow \\ &= \gcd(3+1, 1) \\ &\quad \downarrow \\ &= \gcd(0, 1) \end{aligned}$$

$$m = 6$$

Code. positive nos.

```
int gcd (int A, int B) {  
    if (A == 0) return B;  
  
    if (A % B) {  
        return gcd (A % B, B);  
    } else {  
        return gcd (B, A);  
    }  
}
```

```
int gcd (int A, int B) {  
    if (B == 0) return A;  
  
    return gcd (B, A % B);  
}
```

ensures that
 $A \% B$ is
maintained.

$\text{gcd}(15, 6) \rightarrow \text{gcd}(6, 3)$

$$\begin{array}{c} \downarrow \uparrow 3 \\ \gcd(3, 0) \end{array}$$

$$\gcd(6, 15) \xrightarrow{\leftarrow 3} \gcd(15, 6)$$

$$\begin{array}{c} \downarrow \uparrow 3 \\ \gcd(6, 3) \end{array}$$

$$\begin{array}{c} \downarrow \uparrow 3 \\ \gcd(3, 0) \end{array}$$

$$T.C \rightarrow O(\log(\min(a, b))) \pm \textcircled{5}$$

$$S.C \rightarrow O(\log(\min(a, b))) \pm \textcircled{5}$$

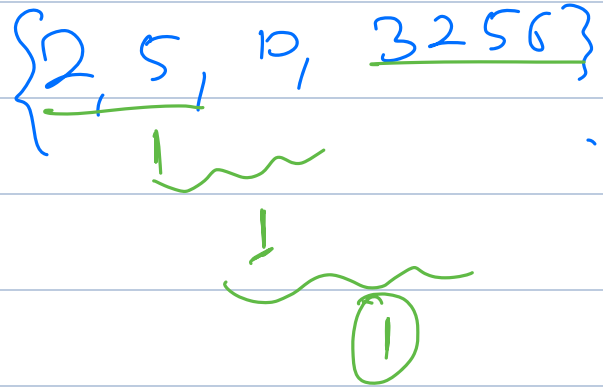
$$\textcircled{5}, \quad \underline{2567983}$$

$$\downarrow \\ \gcd(5, 3)$$

$$\begin{array}{c} 20 \\ \downarrow \\ 10 \\ \downarrow \\ 5 \\ \downarrow \\ 2 \end{array}$$

$$\underline{\log(20)}$$

$$T.C \rightarrow \underline{O(N * \log(\max_i))}$$



Dec 22nd. $\rightarrow$ 7 AM 1st.

23rd to Jan 4th.